

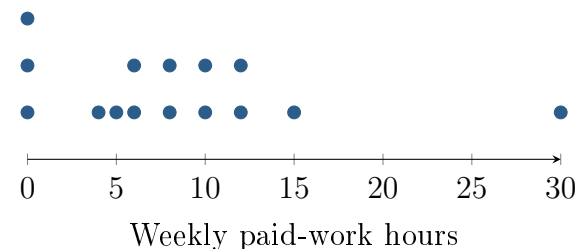
Which Summary Tells the Story?

Unit 1 · Topic 1.7 · THE PROBLEM

Suppose a principal asks, “How much paid work do students do in a week?” Fifteen students report these weekly hours: 0, 0, 0, 4, 5, 6, 6, 8, 8, 10, 10, 12, 12, 15, 30.

Kai: “ $\bar{x} = 8.4$, $s_x \approx 7.5$, and $\text{IQR} = 8$. Report the mean and SD because they use every observation.”

Rosa: “Median = 8, $Q_1 = 4.5$, $Q_3 = 11$, and $\text{IQR} = 6.5$. Report the median and IQR because 30 looks unusual.”



FIRST TAKE — one sentence before you work the parts

Whose report would you send, Kai’s or Rosa’s? Cite one number or dotplot feature.

- DOK 1** (a) State the median and maximum weekly paid-work hours.
- DOK 2** (b) Using the printed quartile rule, find Q_1 , Q_3 , the IQR, and the upper 1.5-IQR fence. Is 30 beyond it? Use Kai’s mean and SD as given.
- DOK 3** (c) Whose IQR follows the printed rule, and how did the other result arise? Choose a corrected summary pair for the principal. Use a dotplot feature to explain why that pair describes typical work hours better than the alternative.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.